

1 Part 4: Understanding Data Sets

2 Summary

3 This Part presents an example of reading in a data set, and ensuring the
4 data is fully understood and properly represented prior to any further
5 analysis.

6 The ability to read in, clean, and understand data sets is a crucial skill.
7 Data do not typically come in a convenient form. Careful consideration
8 should be given to exploring the data to ensure full understanding of the
9 nature of the data set.

1 Example Data Set

2 First, we will read in a data set that we will use in our following examples.

3 Visit the website

4 https://www.sec.gov/opa/data/market-structure/marketstructuredownloadshhtml-by_security.html

5 and download the data from the first quarter of 2017. Create a directory
6 for working on these examples, and place this file in that directory.

7 These data come from the U.S. Securities and Exchange Commission re-
8 spository on market structure. This data set contains daily characteristics
9 of a range of market variables over 5000 equities and ETFs, accumulated
10 over several exchanges.

1 Be sure that the working directory is correctly set in R.

2 Now use the `read.table()` function to read in the file:

```
> fulldata = read.table("q1_2017_all.csv", sep=",",  
+ header=T, quote="")
```

3 This is a “csv” file, i.e., there are “comma-separated-values”. This leads to
4 the use of `sep=“,”`. The `header=T` argument specifies that the first line
5 of the file gives the variable names.

1 **Exercise:** What happens if you use `read.table()` in the example above
2 without `quote=“”`?

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1 Understanding the Variables

2 Whenever a new data set is encountered, one should fully understand its
3 variables. Specific questions to address for each variable:

- 4 1. Describe the information that is stored in this column.
- 5 2. What type of variable is it? (Timestamp, ordinal factor, categorical fac-
6 tor, ratio scale, other?) Be sure that R represents the variable correctly.
- 7 3. Are there any missing values? What is the source of the missingness?
- 8 4. Are there any extreme/inappropriate values? What is the source of
9 the extreme value?

1 Look at the variables:

```
> names(fulldata)
[1] "Date"           "Security"
[3] "Ticker"         "McapRank"
[5] "TurnRank"       "VolatilityRank"
[7] "PriceRank"      "LitVol..000."
[9] "OrderVol..000." "Hidden"
[11] "TradesForHidden" "HiddenVol..000."
[13] "TradeVolForHidden..000." "Cancels"
[15] "LitTrades"      "OddLots"
[17] "TradesForOddLots" "OddLotVol..000."
[19] "TradeVolForOddLots..000."
```

2 Background on the data can be found in the associated README file, avail-
3 able on Canvas in the "Data Sets" folder in the "Files" page. We will refer-
4 ence portions of it below.

1 Column 1: Date

2 The first column in `fulldata` is a date stamp for the observation. R has
3 special functions for handling dates which will prove useful later when
4 working with time series.

5 The `as.Date()` function transforms into the date format:

```
> print(fulldata$Date[1]) # This is Jan. 3, 2017
[1] 20170103

> fulldata$Date = as.Date(as.character(fulldata$Date),
+                          format="%Y%m%d")
```

6 Note that it is necessary to cast the dates as strings, and then specify the
7 format of those strings.

1 Columns 2 and 3: Security and Ticker

2 The variables `Security` and `Ticker` are already appropriately treated as
3 **factors** by R:

```
> is.factor(fulldata$Sec) & is.factor(fulldata$Tick)
[1] TRUE
```

4 We also note that `Security` is either `Stock` or `ETF` (Exchange Traded
5 Fund), and that there are over 5,000 equities under consideration:

```
> levels(fulldata$Security)
[1] "ETF" "Stock"

> nlevels(fulldata$Ticker)
[1] 5338
```

1 **Exercise:** How many of the different ticker symbols are ETFs?

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1 **Exercise:** Execute the commands below, and discuss what you find:

```
> length(unique(fulldata$Date))
> table(table(fulldata$Ticker))
> which.max(table(fulldata$Ticker))
> fulldata[duplicated(fulldata[,c(1,3)]),]
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1 We will remove the duplicated lines using the following command:

```
> fulldata = fulldata[!duplicated(fulldata[,c(1,3)]),
+                        fromLast=TRUE),]
```

2 Note that by using `fromLast=TRUE`, we remove the **first** of each pair of
3 duplicates.

1 **Columns 4 through 7: Rank Variables**

2 The next four columns provide ranks of the stocks/ETFs with respect to
3 key attributes of Market Capitalization, Turnover, Volatility, and Price.

4 In general, we would prefer to start with data in its most “raw” format,
5 and one could construct variables such as these from that data. Alas, we
6 are often left to work with what is available.

- 1 We will change these into **ordered factors** to reflect their ordinal nature:

```
> fulldata$McapRank =
+   factor(fulldata$McapRank, ordered=TRUE)
> fulldata$TurnRank =
+   factor(fulldata$TurnRank, ordered=TRUE)
> fulldata$VolatilityRank =
+   factor(fulldata$VolatilityRank, ordered=TRUE)
> fulldata$PriceRank =
+   factor(fulldata$PriceRank, ordered=TRUE)
```

1 Columns 8 thru 19: The Count Variables

- 2 The remaining column in the data set consist of trade and share counts of
3 different types.
- 4 Note that some of the volume counts are reported in 1000's of shares, hence
5 there are non-integer values among the counts.

- 1 The `summary()` function is a useful first step in understanding the basic
2 properties of the variables. A primary concern is the presence of extreme
3 or impossible values.

```
> summary(fulldata[,8:19])
```

LitVol..000.		OrderVol..000.		Hidden		TradesForHidden	
Min. :	0.00	Min. :	0	Min. :	-143.0	Min. :	0
1st Qu.:	3.46	1st Qu.:	885	1st Qu.:	10.0	1st Qu.:	41
Median :	44.57	Median :	4003	Median :	75.0	Median :	501
Mean :	431.76	Mean :	35747	Mean :	454.4	Mean :	3476
3rd Qu.:	254.41	3rd Qu.:	15581	3rd Qu.:	370.0	3rd Qu.:	2846
Max. :	119014.06	Max. :	7293901	Max. :	89335.0	Max. :	533874

HiddenVol..000.		TradeVolForHidden..000.		Cancels	
Min. :	-7903.64	Min. :	0.00	Min. :	0
1st Qu.:	1.15	1st Qu.:	5.64	1st Qu.:	3126
Median :	8.74	Median :	55.31	Median :	15583
Mean :	66.91	Mean :	498.04	Mean :	73579
3rd Qu.:	44.02	3rd Qu.:	301.58	3rd Qu.:	60112
Max. :	14821.94	Max. :	133714.59	Max. :	7607714

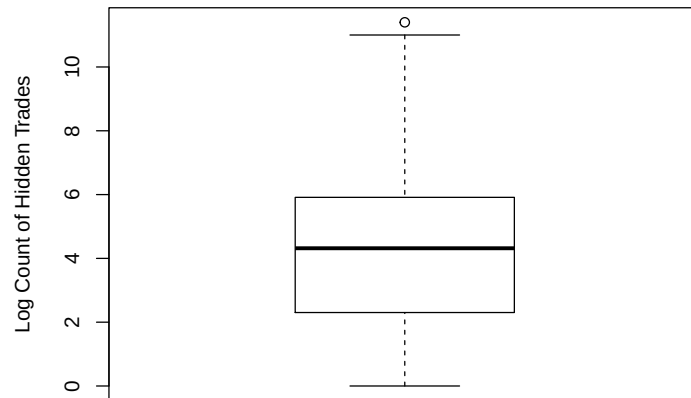
```
LitTrades      OddLots      TradesForOddLots  OddLotVol..000.
Min. : 0      Min. : 0      Min. : 0      Min. : 0.000
1st Qu.: 23    1st Qu.: 8      1st Qu.: 39    1st Qu.: 0.280
Median : 377   Median : 154   Median : 464   Median : 5.254
Mean : 2708   Mean : 916    Mean : 3115   Mean : 32.492
3rd Qu.: 2186  3rd Qu.: 887   3rd Qu.: 2558  3rd Qu.: 31.083
Max. : 383222  Max. : 130106  Max. : 462414  Max. : 3728.885

TradeVolForOddLots..000.
Min. : 0.00
1st Qu.: 5.31
Median : 49.94
Mean : 422.78
3rd Qu.: 260.18
Max. : 108034.79
```

- 1 Of course, graphical tools are useful for understanding the properties of
2 variables. This topic will be covered more fully in the next Part, but here
3 we consider a classic approach: the **boxplot**:

```
> boxplot(log(fulldata$Hidden),  
+         ylab="Log Count of Hidden Trades")
```

- 4 The “box” of the plot extends from the 25th to the 75th percentile of the
5 data, while the solid line in the middle of the box is at the median. The
6 two “arms” extend to the minimum and maximum value, **except** that any
7 value labelled an **outlier** is plotted separately.



2 Missing Data

- 3 Most data sets contain some amount of missing data, for many reasons.

- 1 R uses the special symbol `NA` to represent a missing value.
- 2 As a first step, it is important to recognize how your data set repre-
3 sents missing values, so that R handles them properly. The function
4 `read.table()` has an argument `na.strings` to allow one to specify
5 missing values. **Be careful:** It is not unusual for data sets to use numbers
6 such as `-999` to indicate a missing value.
- 7 In a `.csv` file such as that we are using here, it is common to simply have
8 blank values to indicate missingness, i.e., there are consecutive commas
9 without a value between.

- 1 **Exercise:** Does R appropriately handle cases where a missing value is in-
2 dicated using blank values in a `.csv` file?

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- 1 A useful tool is the function `md.pattern()` that is part of package `mice`.
- 2 This function shows the different “patterns” of missingness in the data set,
- 3 and how often they occur.

- 4 Consider the example on the following page, the result of calling

```
> md.pattern(data.frame(fulldata), plot=FALSE)
```

- 5 This shows that there are 325,082 complete cases in the data set. There is
- 6 one case where `VolatilityRank` is missing, and there are 76 cases where
- 7 `McapRank`, `TurnRank` and `PriceRank` are missing.

```
1      Date Security Ticker LitVol..000. OrderVol..000. Hidden
2 325082      1      1      1      1      1      1
3 76      1      1      1      1      1      1
4 1      1      1      1      1      1      1
5      0      0      0      0      0      0
6      TradesForHidden HiddenVol..000. TradeVolForHidden..000. Cancels
7 325082      1      1      1      1      1
8 76      1      1      1      1      1
9 1      1      1      1      1      1
10      0      0      0      0      0
11      LitTrades OddLots TradesForOddLots OddLotVol..000.
12 325082      1      1      1      1
13 76      1      1      1      1
14 1      1      1      1      1
15      0      0      0      0
16      TradeVolForOddLots..000. VolatilityRank McapRank TurnRank PriceRank
17 325082      1      1      1      1      1
18 76      1      1      0      0      0
19 1      1      0      1      1      1
20      0      1      76      76      76
21
22 325082      0
23 76      3
24 1      1
25      229
```

- 1 **Exercise:** Inspect the output from

```
> fulldata[!complete.cases(fulldata),]
```

- 2 What does this say about the cases with missing values?

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- 1 Of course, more sophisticated visualization of the cases with missing data
- 2 would likely provide more information as to the source of the missingness.
- 3 In the next Part we will consider such tools.